

N100 FINANCIAL INTELLIGENCE

Comprehensive Equity Research Analyst & System Integration Guide

Target Universe:	NIFTY 100 Constituents (92 Active Large/Mid Caps)
Platform Architecture:	Python 3.14, Streamlit, FastAPI REST, SQLite3, ReportLab
Intelligence Modules:	Ratio Engine, Valuation Scorer, NLP Pros/Cons, KMeans Clustering
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Chapter 1: Executive Architecture & Platform Overview

The N100 Financial Intelligence Platform is an institutional-grade equity analytics infrastructure designed for portfolio managers, fundamental research analysts, and risk teams covering Indian equities.

The platform ingests over a decade of audited financial statements (P&L;, Balance Sheet, Cash Flow), market capitalisation multiples, and BSE regulatory filings across 100 constituents, computing 10 core fundamental KPIs, 8 capital allocation archetypes, valuation flags, and NLP thesis points.

Core Technology Stack Components

Layer	Technology	Key Functional Purpose
Data Storage	SQLite3 (B-Tree Indexed)	Stores 10 normalized tables across 100 companies with foreign key integrity.
Analytics Engine	NumPy, Pandas, Scikit-Learn	Computes multi-period CAGR, leverage flags, CFO quality, and KMeans clustering.
User Interface	Streamlit 1.36+ Multi-Page	8 interactive screens with glassmorphic styling and Plotly visualizations.
REST API	FastAPI, Uvicorn, ASGI	16 high-throughput endpoints with CORS and request latency middleware.
Document Engine	ReportLab & Matplotlib	Compiles 2-page company factsheets, 11 sector reports, and portfolio summaries.

Chapter 2: Streamlit Dashboard Navigation

The interactive web dashboard is accessible on <http://localhost:8501> and organized into 8 purpose-built analytical screens:

Screen	Navigation Path	Analyst Workflow & Core Utility
01. Executive Overview	pages/01_home.py	Universe summary KPI cards, fiscal year timeline filter, sector donut chart, quality leaderboard.
02. Company Profile	pages/02_profile.py	Single ticker search, 10-Yr sales/profit bars, ROE/OPM trends, NLP pros/cons badges.
03. Financial Screener	pages/03_screener.py	10 continuous KPI sliders, 6 one-click investment presets, reactive table, CSV export.
04. Peer Comparison	pages/04_peers.py	11 specialized cohorts, 8-axis Plotly Scatterpolar radar chart against peer average.
05. Trend Analysis	pages/05_trends.py	Multi-metric overlay up to 3 financial series across 10 years with point-in-time YoY % annotations.
06. Sector Analysis	pages/06_sectors.py	Macro landscape 4D bubble chart (Revenue vs ROE vs Market Cap) with median bar charts.
07. Capital Allocation	pages/07_capital.py	Hierarchical Plotly Treemap of 8 cash flow archetypes with constituent drilldown.
08. Annual Reports	pages/08_reports.py	Filing repository with live BSE HTTP status verification and direct PDF download links.

Chapter 3: Financial Screener & Preset Strategies

The screener engine evaluates 10 continuous fundamental variables and composite quality rankings:

Preset Strategy	Quantitative Filtering Criteria	Investment Rationale
Quality Compounders	ROE >= 20%, D/E <= 0.5, Rev CAGR >= 10%	Capital-efficient market leaders compounding book value with low financial leverage.
Value Bargains	P/E <= 20x, P/B <= 3x, FCF >= 0	Undervalued opportunities trading below sector median multiples with positive cash flow.
Growth Leaders	Rev CAGR >= 15%, PAT CAGR >= 15%, OPM >= 15%	High top-line and bottom-line compounding with strong operating leverage.
Dividend Yielders	Div Yield >= 2.5%, FCF >= 0, D/E <= 1.0	Generous income distributions backed by healthy free cash flow and conservative debt.
Debt-Free Champions	D/E == 0 (or <= 0.05), ICR >= 10x	Fortress balance sheets immune to rising interest rate cycles.
Turnaround Candidates	PAT CAGR >= 20%, OPM Improving YoY	Operational restructuring candidates demonstrating rapid earnings recovery.

Chapter 4: Peer Group Radar Benchmarking

To eliminate broad-market distortion, the universe is segmented into 11 specialized industry cohorts:

1. Private Banks (Benchmark: HDFCBANK)
2. Public Sector Banks (Benchmark: SBIN)
3. IT Services (Benchmark: TCS)
4. Pharmaceuticals (Benchmark: SUNPHARMA)
5. Automobiles (Benchmark: MARUTI)
6. Life Insurance (Benchmark: HDFCLIFE)
7. Oil & Gas (Benchmark: RELIANCE)
8. Power & Utilities (Benchmark: NTPC)
9. Steel & Mining (Benchmark: TATASTEEL)
10. FMCG (Benchmark: HINDUNILVR)
11. Consumer Finance (Benchmark: BAJFINANCE)

Radar Chart Axes & Normalization

Radar charts project 8 normalized axes: (1) ROE, (2) OPM, (3) NPM, (4) 5-Yr Rev CAGR, (5) 5-Yr PAT CAGR, (6) Asset Turnover, (7) ICR, (8) Composite Quality Score.

Chapter 5: Capital Allocation & Cash Flow Archetypes

Every company's cash flow pattern is evaluated across CFO, CFI, and CFF cash vectors into 8 distinct archetypes:

Cash Vector (CFO, CFI, CFF)	Archetype Label	Institutional Significance
(+, -, -)	Reinvestor / Shareholder Returns	Healthy organic cash generation funding internal growth and paying dividends/debt.
(+, +, -)	Liquidating Assets	Divesting fixed assets or investments to repay financing liabilities.
(-, +, +)	Distress Signal	Operations burning cash; reliant on asset sales and external borrowing.
(-, -, +)	Growth Funded by Debt	High CapEx expansion financed by external capital raises before operational breakeven.
(+, +, +)	Cash Accumulator	Cash inflow across all three activities resulting in large treasury build-up.
(-, -, -)	Pre-Revenue / Cash Burn	Cash drains across operating, investing, and debt repayment.
(+, -, +)	Mixed / Expansionist	Operating cash and new borrowing funding massive strategic CapEx.

Chapter 6: Machine Learning Clustering & Statistics

Unsupervised KMeans clustering (k=5, random_state=42) classifies all 92 companies into data-driven clusters:

- 1. High-Quality Compounders:** Companies with ROE > 25%, OPM > 20%, low leverage (D/E < 0.3).
- 2. Emerging Growth:** High revenue CAGR (>18%) with moderate capital intensity.
- 3. Value Cyclical:** Mature businesses with cyclical operating margins and moderate ROE.
- 4. Defensive Dividend Payers:** Stable utilities/FMCG with steady cash flow and low volatility.
- 5. Distressed or Turnaround:** Higher leverage or depressed profitability requiring structural recovery.

Sector Outlier Detection

Statistical anomalies are flagged when any company metric exhibits absolute Z-score > 3 relative to its broad sector cohort mean and standard deviation.

Chapter 7: Valuation Engine & Relative Flags

The valuation engine establishes sector median P/E benchmarks and identifies relative mispricing:

Overvaluation Caution Flag: Triggered when constituent P/E > 1.5x Sector Median P/E.

Undervaluation Discount Flag: Triggered when constituent P/E < 0.7x Sector Median P/E.

Fair Value Range: Constituent P/E between 0.7x and 1.5x Sector Median P/E.

Free Cash Flow Yield

$$\text{FCF Yield (\%)} = \frac{\text{Free Cash Flow (Cr)}}{\text{Market Capitalisation (Cr)}} \times 100$$

Chapter 8: FastAPI REST Service & Integration

The API service runs on port 8000. Key cURL command examples:

Endpoint & Purpose	Example cURL Request
Health Check	curl http://localhost:8000/api/v1/health
Company Profile	curl http://localhost:8000/api/v1/companies/TCS
P&L; History	curl http://localhost:8000/api/v1/companies/TCS/pl?from_year=2020
Screener Query	curl http://localhost:8000/api/v1/screener?min_roe=15&max_de=1
Sector Medians	curl http://localhost:8000/api/v1/sectors
Peer Radar Data	curl http://localhost:8000/api/v1/companies/TCS/peers/compare
Download Tearsheet	curl -O http://localhost:8000/api/v1/companies/TCS/tearsheet

Chapter 9: Data Quality Rules & System Maintenance

16 Automated Data Quality Rules are enforced across all 12 ingested datasets:

DQ-01: Primary key null checks.

DQ-02: Primary key uniqueness.

DQ-03: Foreign key referential integrity.

DQ-04: Mandatory company attributes.

DQ-05: Year range bounds [2000, 2030].

DQ-06 to DQ-09: Stock price non-negativity, High-Low bounds, ISO date validation.

DQ-10 to DQ-11: Balance sheet accounting equation ($\text{Assets} == \text{Liabilities}$) and sub-component reconciliation.

DQ-12: Cash flow waterfall reconciliation ($\text{CFO} + \text{CFI} + \text{CFF} == \text{Net Cash Flow}$).

DQ-13 to DQ-16: Valuation non-negativity, sector weights, URL protocols, and composite entity-time uniqueness.

Troubleshooting Tips

1. If a company ticker shows missing ratios, verify `output/validation_failures.csv` for data anomalies.
2. If port 8000 is occupied, run FastAPI on custom port: `uvicorn src.api.main:app --port 8080`.
3. To re-run test suite with HTML report: `pytest --html=reports/pytest_report.html --self-contained-html`.